

EE6203 Computer Controlled Systems

Weeks 1–4 Review Notes

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Weekly coverage map

Week	Topics	PDF pages
1	Feedback, digital-control architecture, signals, z -transform pairs/properties, poles/zeros, initial/final values	1–48
2	Inverse z methods, difference equations, impulse sampling, starred transforms, ZOH	49–88
Self-study	First-order hold (optional)	89–93
3	Pulse transfer functions, sampled block diagrams, ZOH/plant, controller and PID models	94–143
4	s - z mapping, stability, transient/steady-state response, disturbances	144–195

Weeks 1–3 are verified from the supplied transcripts. Week 2 explicitly hands pulse transfer functions to the next lecture. The Week 3 transcript completes Chapter 3, starts Chapter 4, and confirms that Quiz 1 covers Chapters 1–3 only.

Additional practice sources: [resources/course-exercises.pdf](#) (5 pages) and [resources/course-exercises-solutions.pdf](#) (26 pages). The solution compilation retains some older EE4273 headings and handwriting, so incorporated calculations are independently re-derived.

1 Digital-control essentials

Week 1

Unity feedback: $e(t) = r(t) - y(t)$, where t is time, $r(t)$ is the reference or setpoint, $y(t)$ is the measured output, and $e(t)$ is the tracking error. The signals r , y , and e have the same physical unit. Main requirements: stability, transient response, steady-state accuracy, and possibly optimality.

Design vs. analysis: design chooses a controller from the plant and specifications; analysis verifies the closed-loop stability and performance.

Periodic sampling:

$$t_k = kT, \quad \omega_s = \frac{2\pi}{T}, \quad x[k] = x(kT).$$

Sampled-data means discrete time with continuous amplitude; digital means discrete time and quantized amplitude. ADC sampling and amplitude quantization are distinct. Typical path:

$$r \rightarrow \text{sum} \rightarrow \text{ADC} \rightarrow \text{computer} \rightarrow \text{DAC+ZOH} \rightarrow \text{plant} \rightarrow y.$$

Checklist: identify every sampler and hold before simplifying a block diagram.

[PDF pp. 7–20]

2 The z transform

Week 1 through final values; Week 2 from inverse methods onward

2.1 Definitions and pairs

One-sided causal transform and power-series interpretation:

$$X(z) = \mathcal{Z}\{x[k]\} = \sum_{k=0}^{\infty} x[k]z^{-k} = x[0] + x[1]z^{-1} + \dots$$

$x[k]$	$X(z)$
$\delta[k - n]$	z^{-n}
$\mathbf{1}[k]$	$(1 - z^{-1})^{-1} = z/(z - 1)$
$k\mathbf{1}[k]$	$z^{-1}/(1 - z^{-1})^2 = z/(z - 1)^2$
$a^k\mathbf{1}[k]$	$(1 - az^{-1})^{-1} = z/(z - a)$
$\sin(\omega kT)\mathbf{1}[k]$	$z \sin(\omega T)/(z^2 - 2z \cos \omega T + 1)$
$\cos(\omega kT)\mathbf{1}[k]$	$(z^2 - z \cos \omega T)/(z^2 - 2z \cos \omega T + 1)$

In the last two rows, every ω in a trigonometric argument is multiplied by T .

2.2 Properties

$$\begin{aligned}\mathcal{Z}\{\alpha x[k] + \beta y[k]\} &= \alpha X(z) + \beta Y(z), \\ \mathcal{Z}\{a^k x[k]\} &= X(z/a), \\ \mathcal{Z}\{x[k-n] \mathbf{1}[k-n]\} &= z^{-n} X(z), \\ \mathcal{Z}\{x[k+n]\} &= z^n X(z) - \sum_{i=0}^{n-1} z^{n-i} x[i], \\ \mathcal{Z}\{e^{-akT} x[k]\} &= X(ze^{aT}).\end{aligned}$$

Initial value:

$$x[0] = \lim_{z \rightarrow \infty} X(z).$$

Final value:

$$x[\infty] = \lim_{z \rightarrow 1} (1 - z^{-1}) X(z),$$

valid only if all poles of $(1 - z^{-1})X(z)$ lie strictly inside $|z| = 1$.

A simple pole of $X(z)$ at $z = 1$ may represent a finite constant. A repeated pole there grows, and other unit-circle poles oscillate; neither case has a final value.

2.3 Inverse methods

Week 2 begins

- Long division in z^{-1} : read off samples; usually no closed form.
- Recurrence computation: apply $\delta[k]$ to the difference equation; efficient for a sample table.
- Partial fractions: expand $X(z)/z$; simple $1/(z - p)$ terms give p^k and a closed form.
- Repeated pole: expect kp^{k-1} factors.
- Complex pair: combine into real damped sine/cosine terms.
- Residues: $x[k] = (2\pi j)^{-1} \oint X(z) z^{k-1} dz$.

Inverse z transformation determines $x[k]$, not a unique continuous $x(t)$; reconstruction between samples needs an extra assumption.

For

$$x[k+2] + 3x[k+1] + 2x[k] = 0, \quad x[0] = 0, \quad x[1] = 1,$$

the transform is $X(z) = z/[(z+1)(z+2)]$ and

$$x[k] = (-1)^k - (-2)^k.$$

Common mistakes: treating an advance as a pure z^n multiplier; using the final-value theorem on an oscillatory/unstable sequence; doing PFE on the wrong form; ignoring the ROC. [PDF pp. 21–70]

3 Sampled-data modeling

Week 2 through the ZOH; Week 3 from pulse transfer functions onward

3.1 Sampler and holds

Ideal impulse sampler:

$$x^*(t) = \sum_{k=0}^{\infty} x(kT) \delta(t - kT), \quad X^*(s) = \sum_{k=0}^{\infty} x(kT) e^{-kTs}.$$

With $z = e^{sT}$, $X^*(s) = X(z)$.

The coefficient $x(kT)$ is an impulse's area (strength), not a finite height. Also, $z = e^{sT}$ links $X^*(s)$ to $X(z)$; it does not justify substituting into an ordinary $X(s)$ or $G_p(s)$.

Zero-order hold:

$$h(kT + \tau) = x(kT), \quad 0 \leq \tau < T, \quad G_{h0}(s) = \frac{1 - e^{-sT}}{s}.$$

Equivalently, the ZOH is a sum of one-period rectangular windows; their Laplace transform produces $(1 - e^{-sT})/s$. First-order hold (optional Week 2 self-study):

$$G_{h1}(s) = \left(\frac{1 - e^{-sT}}{s} \right)^2 \frac{Ts + 1}{T}.$$

3.2 Pulse transfer function

Week 3 begins

At sampling instants, continuous convolution becomes

$$y(kT) = \sum_{h=0}^k g((k-h)T)x(hT),$$

so

$$Y(z) = G(z)X(z), \quad G(z) = \mathcal{Z}\{g(kT)\}.$$

Input-sampler rule:

- sampler before $G(s)$: $Y(z) = G(z)X(z)$;
- no input sampler: $Y(z) = \mathcal{Z}\{G(s)X(s)\}$, generally not $G(z)X(z)$;
- sampler between cascaded G, H : pulse TF $G(z)H(z)$;
- no intermediate sampler: pulse TF $GH(z) = \mathcal{Z}\{G(s)H(s)\} \neq G(z)H(z)$.

ZOH plus plant:

$$G(z) = \mathcal{Z} \left\{ \frac{1 - e^{-sT}}{s} G_p(s) \right\} = (1 - z^{-1}) \mathcal{Z} \left\{ \frac{G_p(s)}{s} \right\}.$$

3.3 Digital controller and PID

Controller recurrence

$$m[k] + \sum_{i=1}^n a_i m[k-i] = \sum_{j=0}^m b_j e[k-j]$$

gives

$$G_D(z) = \frac{M(z)}{E(z)} = \frac{\sum_{j=0}^m b_j z^{-j}}{1 + \sum_{i=1}^n a_i z^{-i}}.$$

Unity-feedback sampled loop:

$$\frac{C(z)}{R(z)} = \frac{G_D(z)G(z)}{1 + G_D(z)G(z)}.$$

Trapezoidal-integral/backward-difference PID:

$$G_D(z) = K_P + \frac{K_I}{1 - z^{-1}} + K_D(1 - z^{-1}),$$

$$K_P = K - \frac{KT}{2T_i}, \quad K_I = \frac{KT}{T_i}, \quad K_D = \frac{KT_d}{T}.$$

Common mistakes: substituting $z = e^{sT}$ directly into $G_p(s)$; moving or deleting samplers while reducing a diagram; forgetting that $G(z)H(z) \neq GH(z)$. [PDF pp. 71–143]

4 s -plane to z -plane mapping

Chapter 4: retained for later course study; not assessed in Quiz 1.

Week 4

For $s = \sigma + j\omega$,

$$z = e^{sT} = e^{\sigma T} e^{j\omega T}, \quad |z| = e^{\sigma T}, \quad \arg z = \omega T \pmod{2\pi}.$$

s -plane object	z -plane image
$\sigma < 0$	$ z < 1$
$\sigma = 0$	unit circle
$\sigma = \sigma_0$	circle $ z = e^{\sigma_0 T}$
$\omega = \omega_0$	radial line $\arg z = \omega_0 T$
primary strip	$-\pi/T \leq \omega \leq \pi/T$ maps once around

For $s = -\zeta\omega_n \pm j\omega_d$,

$$|z| = e^{-\zeta\omega_n T}, \quad \arg z = \pm\omega_d T.$$

With the 2% approximation $t_s \approx 4/(\zeta\omega_n)$,

$$t_s \leq t_s^{\max} \Rightarrow |z| \leq \exp(-4T/t_s^{\max}).$$

Constant damping-ratio rays map to logarithmic spirals. Frequencies separated by $2\pi/T$ map to the same angle.
[PDF pp. 144–161]

5 Discrete-time stability

Week 4

Closed-loop asymptotic stability: all roots of $P(z)$ strictly inside $|z| = 1$. Simple unit-circle poles are marginal; repeated unit-circle poles are unstable.

For

$$P(z) = a_0 z^n + a_1 z^{n-1} + \cdots + a_n, \quad a_0 > 0,$$

the first Jury conditions are

$$|a_n| < a_0, \quad P(1) > 0, \quad (-1)^n P(-1) > 0.$$

They are sufficient for $n = 2$. For higher order, continue the Jury table with alternating forward/reversed rows and require successive leading/trailing magnitude inequalities.

Example from the source:

$$P(z) = z^2 + (0.3679K - 1.3679)z + (0.3679 + 0.2642K)$$

is stable for

$$0 < K < 2.3925.$$

Bilinear route:

$$z = \frac{w+1}{w-1}, \quad w = \frac{z+1}{z-1}.$$

This maps $|z| < 1$ to $\Re(w) < 0$. Substitute, clear the denominator, and apply Routh. Jury is usually shorter; Routh also counts unstable roots.
[PDF pp. 162–179]

6 Transient and steady-state response

Week 4

Step-response terms: delay time t_d , rise time t_r , peak time t_p , maximum overshoot

$$M_p = \frac{c(t_p) - c(\infty)}{c(\infty)} 100\%,$$

and settling time t_s (usually a 2% band).

For the standard loop,

$$E(z) = \frac{R(z)}{1 + G(z)H(z)}, \quad e_{ss} = \lim_{z \rightarrow 1} (1 - z^{-1})E(z).$$

Static error constants:

$$\begin{aligned} K_p &= \lim_{z \rightarrow 1} GH, & e_{ss, \text{step}} &= \frac{1}{1 + K_p}, \\ K_v &= \lim_{z \rightarrow 1} \frac{(1 - z^{-1})GH}{T}, & e_{ss, \text{ramp}} &= \frac{1}{K_v}, \\ K_a &= \lim_{z \rightarrow 1} \frac{(1 - z^{-1})^2 GH}{T^2}, & e_{ss, \text{accel}} &= \frac{1}{K_a}. \end{aligned}$$

Type	Step	Ramp	Acceleration
0	$1/(1 + K_p)$	∞	∞
1	0	$1/K_v$	∞
2	0	0	$1/K_a$

System type is the number of uncanceled open-loop poles at $z = 1$. The table assumes a stable loop and the lecture's standard configuration. Actuating error $R - B$ need not equal tracking error $R - C$.

For the lecture disturbance configuration,

$$E(z) = -\frac{G(z)}{1 + G_D(z)G(z)}N(z).$$

Large loop gain reduces error. An uncanceled controller pole at $z = 1$, with no plant zero there, gives zero steady-state error to a constant disturbance if the closed loop is stable. [PDF pp. 180–195]

7 Exam workflow

1. Draw the sampled-data topology and label T .
2. Convert ZOH plus plant with $(1 - z^{-1})\mathcal{Z}\{G_p(s)/s\}$.
3. Preserve samplers when combining blocks.
4. Form the characteristic polynomial; apply Jury or bilinear/Routh.
5. Map performance regions using both $|z| = e^{\sigma T}$ and $\arg z = \omega T$.
6. For steady-state error, verify stability, input transform, error definition, and final-value conditions.
7. Check the first few samples or substitute a boundary gain to catch algebra errors.

The source recap on PDF pages 196–225 repeats these four chapters and adds no distinct topic.

A Quiz 1 closed-book review

Confirmed format and scope: 30 minutes; Chapters 1–3 only; closed book; calculator permitted; necessary transform pairs supplied; show working for non-multiple-choice questions. Unclear transcript details about the exact time, seating, and absence arrangements are omitted.

The two unofficial recalled versions differ, but together emphasize direct z -transform construction, one-sided difference equations, final-value conditions, sampler/ZOH modeling, and pulse-transfer topology. Use the following as a memory sheet; the lecture PDF and Week 3 announcement remain authoritative. Chapter 4 mapping, Jury/bilinear stability, transient response, system type, and steady-state-error tables are not Quiz 1 material.

A.1 Core formulas to reconstruct

Definition and pairs

$$\begin{aligned} X(z) &= \sum_{k=0}^{\infty} x[k]z^{-k}, \\ \mathcal{Z}\{\delta[k-n]\} &= z^{-n}, \\ \mathcal{Z}\{\mathbf{1}[k]\} &= \frac{1}{1-z^{-1}}, \\ \mathcal{Z}\{a^k \mathbf{1}[k]\} &= \frac{1}{1-az^{-1}}, \\ \mathcal{Z}\{k \mathbf{1}[k]\} &= \frac{z^{-1}}{(1-z^{-1})^2}. \end{aligned}$$

One-sided shifts and values

$$\begin{aligned} \mathcal{Z}\{x[k-n]\mathbf{1}[k-n]\} &= z^{-n}X(z), \\ \mathcal{Z}\{x[k+1]\} &= zX(z) - zx[0], \\ \mathcal{Z}\{x[k+2]\} &= z^2X(z) - z^2x[0] - zx[1], \\ x[0] &= \lim_{z \rightarrow \infty} X(z), \\ x[\infty] &= \lim_{z \rightarrow 1} (1-z^{-1})X(z). \end{aligned}$$

Final value is valid iff every pole of $(1-z^{-1})X(z)$ is strictly inside $|z| = 1$. One simple pole of X at $z = 1$ is allowed; a repeated pole there, another unit-circle pole, or any outside pole is not.

A.2 Inverse-transform decision

Need	Fastest method
First few samples	Long division in z^{-1} or direct recurrence.
Closed form	Partial fractions of $X(z)/z$; pair $1/(z-a)$ with a^k .
Repeated/complex poles	Include polynomial factors in k ; combine conjugates into real sine/cosine terms.
Theory/general inversion	Contour integral and residues.

A.3 Sampling, ZOH, and topology

$$\begin{aligned} x^*(t) &= \sum_{k=0}^{\infty} x(kT)\delta(t-kT), \\ X^*(s) &= \sum_{k=0}^{\infty} x(kT)e^{-kTs}, \\ G_{h0}(s) &= \frac{1-e^{-sT}}{s}, \end{aligned} \quad \begin{aligned} z &= e^{sT}, \\ G(z) &= (1-z^{-1})\mathcal{Z}\left\{\frac{G_p(s)}{s}\right\}. \end{aligned}$$

- Input sampler present: $Y(z) = G(z)X(z)$.
- No input sampler: $Y(z) = \mathcal{Z}\{G(s)X(s)\}$; do not separate the factors.
- Intermediate sampler: cascades reduce to $G(z)H(z)$.
- No intermediate sampler: use $GH(z) = \mathcal{Z}\{G(s)H(s)\} \neq G(z)H(z)$.
- A pulse transfer function $C(z)/R(z)$ does not exist if the external input remains inseparably inside a starred continuous-time product.

A.4 Exercise-backed checks

For a parameter recurrence

$$y[k+1] - \beta y[k] = \mathbf{1}[k], \quad y[0] = 0,$$

$$Y(z) = \frac{z}{(z-1)(z-\beta)}, \quad y[k] = \frac{1-\beta^k}{1-\beta} \quad (\beta \neq 1).$$

After cancelling the permitted simple pole at $z = 1$, inspect $z = \beta$: the final-value theorem is valid exactly for $|\beta| < 1$. $\beta = 1$ grows, $\beta = -1$ oscillates, and $|\beta| > 1$ diverges.

For a controller recurrence with zero initial conditions, collect present and delayed controller/output terms before closing the loop:

$$m[k] + \sum_{i=1}^n a_i m[k-i] = \sum_{j=0}^r b_j e[k-j] \implies G_D(z) = \frac{\sum_{j=0}^r b_j z^{-j}}{1 + \sum_{i=1}^n a_i z^{-i}}.$$

Reduce sampler-ZOH-plant topology to $G(z)$ *first*; then use $G_D(z)G(z)$ only when the resulting sampled loop permits it.

Useful supplied-exercise patterns:

- $\mathcal{Z}\{kx[k]\} = -z \, dX/dz$; apply it repeatedly for powers such as k^3 .
- A repeated pole $(1 - az^{-1})^{-2}$ produces a factor ka^{k-1} .
- A factor z^{-n} delays a causal sequence by n samples and inserts n leading zeros.
- Deriving a pulse transfer through a transform table and through inverse Laplace plus sampling must give the same result.

A.5 Sampled feedback and PID

For a correctly reduced unity negative-feedback loop,

$$\frac{C(z)}{R(z)} = \frac{G_D(z)G(z)}{1 + G_D(z)G(z)}.$$

For the course's digital PID realization,

$$G_D(z) = K_P + \frac{K_I}{1 - z^{-1}} + K_D(1 - z^{-1}).$$

Do not identify an arbitrary controller recurrence as PID until it has been rearranged into the stated realization.

With $T = 1$, a ZOH and plant K/s reduce to

$$G(z) = \frac{Kz^{-1}}{1 - z^{-1}}.$$

For a unit-step reference in unity feedback,

$$C(z) = \frac{Kz^{-1}}{[1 - (1 - K)z^{-1}](1 - z^{-1})}, \quad c[k] = 1 - (1 - K)^k.$$

The samples converge to one for $0 < K < 2$ because $|1 - K| < 1$. This is a Chapter 3 pulse-transfer/pole check, not a Chapter 4 Jury calculation.

A.6 Thirty-second setup checklist

1. Label T , every sampler, every hold, and every continuous/sampled signal.
2. For piecewise data, write the sample table and begin with the defining sum.
3. For a recurrence, write the initial-condition terms before collecting $Y(z)$.
4. For a ZOH/plant, transform $G_p(s)/s$ and multiply by $(1 - z^{-1})$.
5. Before final value, cancel only algebraically and inspect the remaining poles.
6. If a parameter controls a pole, split the answer into parameter ranges before taking a limit.
7. For a controller recurrence, obtain $G_D(z)$ before forming any closed-loop fraction.
8. Check by first samples, recurrence balance, DC gain, or a boundary pole.

A.7 Six-day drill route

Day	Two-hour focus
1	Definition, transform-pair derivations, finite and piecewise sums.
2	One-sided shifts, difference equations, inverse methods, final value, and parameter-pole cases.
3	$x(t)$ versus $x[k]$, impulse sampling, starred and z transforms.
4	ZOH/plant conversion, controller recurrences, pulse-transfer topology, open/closed loops.
5	Repeated-pole and delayed inversions; controller/PID recurrences; supplied Exercise groups 2–4.
6	Formula dump, two 30-minute mocks, marking, and error repair.

Use the worked methods, archetypes, drills, solutions, and two mocks in Appendix A of *quiz1-prep-learning.pdf*.
Target at least 80% on the second mock with no repeated conceptual error. [PDF pp. 21–143]